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PROFESSIONAL SUMMARY

Computer Science & Engineering undergraduate (CGPA: 8.26) with hands-on experience across backend, full-stack, and applied ML systems, backed by strong fundamentals in **data structures, algorithms, OOP, DBMS, operating systems, and computer networks**. Shipped **5+ merged pull requests** to a live production codebase as a technical intern, and led a **Top 10 (of 72 teams)** hackathon finish. Seeking a **Software Development Engineer / Associate Software Engineer** role.

EDUCATION

Canara Engineering College, Mangalore

B.E in Computer Science & Engineering

CGPA: 8.26

2026

KSEAB (SSLC/PUC)

SSLC: 88.96% — PUC: 92.96%

EXPERIENCE

Orbit — Intern, Technical Team Member

Feb 2026 – Present

- Developed a location-based social discovery platform, working within a monorepo with a React Native/Expo mobile client and a Node.js/Express REST API backend.
- Engineered real-time chat infrastructure using WebSockets and geolocation-based event discovery services powering the platform's core social features.
- Deployed and maintained the backend in production on Render with Sentry error monitoring and Docker-based environment consistency; shipped mobile builds via Expo Application Services (EAS).

ACHIEVEMENTS & LEADERSHIP

Core Unleashed Hackathon — Team Lead, “Digital Sentinels”

Apr 2025

- Ranked among the **Top 10 teams** out of **72 teams** and pitched our solution to the final judging panel after advancing through multiple evaluation rounds.
- Led a team to build PII Shield, a privacy-first PII redaction tool, in a **36-hour hackathon**, driving the architectural decision to run **OCR (Tesseract)** entirely locally to eliminate cloud dependency and guarantee zero data retention.
- Engineered a stateless backend with a deterministic finding-ID system and a **3-tier bounding-box** fallback (exact character → line-level → sidebar-only) to keep redaction accurate despite imperfect OCR spatial mapping.

Feature Development & Deployment

Ongoing

- **Merged 5+ pull requests** into Orbit's production codebase, shipping UI improvements, new features, and backend fixes.
- Redesigned the landing page and signup flow UI, and built a **location-picker dropdown that reverse-geocodes** a user-dropped map pin into an address.
- Implemented **event creation with feed-level visibility**, enabling public events to surface across all users' feeds.
- Resolved **API rate-limiting** issues and deployed the backend via Docker; configured EAS builds for the React Native app.

PROJECTS

GraphSentinel — Graph Neural Network Cyber Defense System

Python, PyTorch Geometric, GraphSAGE

- Designed and trained a GraphSAGE-based intrusion detection model on CICIDS-2017, choosing an inductive architecture over GCN specifically to classify previously unseen IPs at inference time.
- Engineered a **7-feature behavioral signal set** (port entropy, byte asymmetry, out-degree, etc.) from raw traffic data, and resolved severe class imbalance (128K vs. 1,956 samples) using hybrid under-sampling with SMOTE and weighted cross-entropy loss.
- Achieved **99.57% accuracy** and **0.9975 AUC-ROC**, with inference optimized to **~120ms per graph** to fit within a 5-second real-time detection budget.

HVSapp — Healthcare Handoff Validation Platform

FastAPI, React Native, PostgreSQL, Celery

- Built the backend for a clinical handoff and medication tracking system, decoupling an **ASR transcription pipeline into async Celery task queues** to avoid blocking real-time request handling.
- Implemented PHI field encryption and medication safety tracking via Alembic migrations, structuring the schema around clear domain boundaries (auth, medication, patients, transcription).
- Applied targeted database indexing and async task decoupling to improve API responsiveness under concurrent clinical workflows.

TECHNICAL SKILLS

• **Core CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

• **Languages:** Java, Python, JavaScript, SQL

• **Backend & Cloud:** Node.js, Express, FastAPI, Flask, AWS (EC2, S3, VPC), MongoDB, PostgreSQL, Redis

• **Frontend & Mobile:** React.js, React Native, Redux, Tailwind CSS

• **ML & Data:** PyTorch, PyTorch Geometric (GraphSAGE), Scikit-learn, OpenCV, MediaPipe

• **DevOps & Tools:** Git/GitHub, Docker, WebSockets, Celery, CI/CD basics, Linux Administration